THE STAR OPERATION COMMUTES WITH PASSING TO DIRECT SUMMANDS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 10966-10973, following the statement at 10955-10964.

Editorial note. No omissions: the witnessing triangle is displayed and X is read off as a direct summand.

Allowed standard prerequisites (editorial).

- Star of full subcategories. definition of A star B and smd earlier in the section (same chapter). Supplies the two operations being compared.

Lemma 0.1. *Let $\mathcal{T}$ be a triangulated category. Given full subcategories $\mathcal{A}$, $\mathcal{B}$ we have $smd(\mathcal{A}) \star smd(\mathcal{B}) \subset smd(\mathcal{A} \star \mathcal{B})$ and $smd(smd(\mathcal{A}) \star smd(\mathcal{B})) = smd(\mathcal{A} \star \mathcal{B})$.*

Proof. Suppose we have a distinguished triangle $A_1 \to X \to B_1$ where $A_1 \oplus A_2 \in \mathrm{Ob}(\mathcal{A})$ and $B_1 \oplus B_2 \in \mathrm{Ob}(\mathcal{B})$. Then we obtain a distinguished triangle $A_1 \oplus A_2 \to A_2 \oplus X \oplus B_2 \to B_1 \oplus B_2$ which proves that X is in $smd(\mathcal{A} \star \mathcal{B})$. This proves the inclusion. The equality follows trivially from this. $\square$

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